

SENSORY SCHOOL FOR AUTISM

CAPSTONE PROJECT

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Interior Design degree

By

V.KSHIRAJA (42102008)



DEPARTMENT OF DESIGN

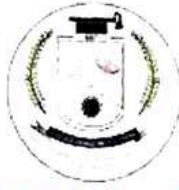
SCHOOL OF BUILDING AND ENVIRONMENT

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**INSTITUTE OF SCIENCE AND TECHNOLOGY
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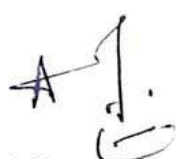
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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of **V. KSHIRAJA (42102008)** who carried out the project entitled **Sensory School For Autism** under our supervision from December 2025 to May 2026


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ACKNOWLEDGEMENT

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ABSTRACT

This thesis focuses on designing an autism-friendly school for children with Autism Spectrum Disorder (ASD). The main aim is to create a safe, comfortable, and supportive environment that helps children learn, communicate, and develop better in their daily activities. Children with autism often have different sensory needs, such as sensitivity to light, sound, texture, and colors. This design considers both hypersensitivity and hyposensitivity by using soft lighting, controlled acoustics, calming color palettes, and simple materials. The spaces are planned with clear zoning, where active and quiet areas are separated to reduce stress and confusion.

The school design includes classrooms, therapy rooms, sensory rooms, quiet spaces, and play areas. Special attention is given to circulation and wayfinding so that children can easily understand and move through the space independently. Flexible furniture and safe materials are used to ensure comfort and safety.

Natural light, ventilation, and elements like greenery are included to create a healthy and positive atmosphere. Overall, this thesis shows how thoughtful interior design can improve the learning experience, behavior, and well-being of children with autism, helping them feel more confident and independent.

This thesis also highlights the importance of inclusive design in educational spaces. It aims to create an environment that not only supports children with autism but also encourages interaction, social development, and a sense of belonging. The design promotes flexibility to adapt to different learning needs and behaviors.

By combining functionality with sensitivity to user needs, the project creates a balanced space that supports both learning and emotional growth. This approach can be used as a reference for future educational designs that focus on special needs and inclusive environments.

1.INTRODUCTION

1.1 AIM

- To design an autism-friendly school environment
- To create a safe, comfortable, and supportive space
- To address sensory and behavioral needs of children with ASD
- To improve learning, independence, and well-being

1.2 OBJECTIVES

To study sensory needs (light, sound, color, texture)

- To reduce stress using proper lighting and acoustics
- To create clear zoning (active, quiet, therapy areas)
- To design simple and easy circulation
- To improve wayfinding for better navigation
- To provide flexible spaces for learning and activities
- To use safe, non-toxic, and durable materials
- To include natural light, ventilation, and greenery
- To support social interaction and independence

1.3 SCOPE

- Focus on interior design of an autism school
- Design classrooms, therapy rooms, sensory rooms, play areas
- Apply sensory design principles
- Improve comfort through color, lighting, and materials
- Create functional and user-friendly layouts
- Can be used as a reference for future special schools

1.4 LIMITATIONS

- Project is theoretical (not executed in real life)
- Budget limitations may affect design choices
- Not all autistic children have the same needs

- Limited real user interaction or feedback
- Site and climate factors may be limited

1.5 ABOUT AUTISM SCHOOL

- **Definition:** An autism school is a specialized educational space designed for children with Autism Spectrum Disorder (ASD), focusing on their unique sensory and learning needs.
- **Learning Environment:** These schools provide a structured and supportive setting that helps children improve communication, social interaction, and learning abilities through guided and personalized methods.
- **Sensitivity to Surroundings:** Children with autism are highly sensitive to environmental factors such as light, sound, and space. Design aims to reduce stress and discomfort Helps in improving focus and behavior
- **Design Approach:** Spaces are carefully planned to create a calm and comfortable environment. Use of soft and diffused lighting Reduction of noise through acoustic materials Use of soothing and neutral color palettes
- **Special Spaces:** Autism schools include different types of spaces to support various needs Therapy rooms for speech and behavioral development Sensory rooms for stimulation and relaxation Quiet zones for calming and stress relief Play areas for physical and social activities
- **Overall Purpose:** These schools are not only for education but also focus on Emotional support Behavioral improvement Development of independence and confidence

1.6 HISTORICAL SIGNIFICANCE

- Autism identified in the early 20th century
- Earlier, lack of awareness and proper education systems
- Development of special education schools over time
- Increased focus on inclusive and sensory-friendly design
- Modern schools support comfort, learning, and behavior
- Shows importance of inclusive design in architecture

2. LITERATURE STUDY

2.1 DESIGN NORMS AND STANDARDS

- **Sensory Control:** Spaces should be designed to reduce sensory overload. Control light, sound, color, and texture Avoid overstimulation and distractions Provide a calm and balanced environment
- **Lighting Design:** Lighting should be soft and comfortable for children. Use natural daylight wherever possible Avoid harsh and flickering lights Provide diffused and glare-free artificial lighting
- **Acoustic Control:** Noise levels must be minimized for better concentration Use sound-absorbing materials (panels, carpets) Avoid echo and external noise Provide quiet zones for relaxation
- **Spatial Zoning:** Spaces should be clearly divided based on activities. Active zones (play, group learning) Quiet zones (rest, individual work) Therapy zones (focused activities) Helps in reducing confusion and stress
- **Color Scheme:** Colors should create a calm and comfortable environment. Use soft, neutral, and pastel shades Avoid bright, contrasting, and bold colors Maintain visual balance
- **Material Selection:** Materials should be safe and user-friendly. Use non-toxic and durable materials Prefer matte finishes over glossy surfaces Easy to clean and maintain
- **Connection with Nature:** Incorporate natural elements for well-being. Natural light and ventilation Use of greenery and outdoor spaces Creates a calming and healthy environment

2.2 DATA COLLECTION

The data collection for this study focuses on understanding the design requirements of autism schools and the needs of children with Autism Spectrum Disorder (ASD). Literature review of books, research papers, and online sources provided information about sensory design, spatial planning, and design standards. Case references of similar autism centers helped in understanding space planning and user requirements. Surveys and discussions with teachers and therapists helped to understand user behavior, daily activities, and challenges faced by children. Site visits and observations were also used to study real-life environments, movement patterns, and space usage.

SENSORY SCHOOL FOR AUTISM



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3.SPECIAL STUDY AND THRUST AREA

3.1 FURNITURE :

- Furniture design is very important in an autism-friendly school as it directly affects safety, comfort, and usability. All furniture should have rounded edges and smooth finishes to avoid injuries.
- Materials used should be non-toxic, durable, and easy to clean. The size and height of furniture must be suitable for children, ensuring proper posture and comfort during activities.
- Flexible and movable furniture is preferred so that spaces can be easily rearranged for individual learning or group activities. Storage units should be simple and easily accessible to help children develop independence.
- Soft seating options like bean bags or cushioned chairs can also be provided in quiet areas to create a relaxed environment

3.2 LIGHTING :

- Lighting plays a major role in controlling the sensory experience of children with autism. Natural light should be maximized as it creates a healthy and pleasant environment. Artificial lighting should be soft, diffused, and evenly distributed to avoid harsh shadows and glare.
- Flickering lights, especially fluorescent lighting, should be avoided as they may cause discomfort and distraction. Adjustable lighting options can be provided to suit different activities and individual needs.
- The use of warm or neutral color temperature lighting helps in creating a calm and stress-free atmosphere, improving focus and reducing anxiety.

3.3 FLOORING MATERIAL

- Flooring selection is important for safety, comfort, and acoustic control. Non-slip flooring is essential to prevent accidents, especially in active areas. Soft and cushioned materials help in reducing impact and are safer for children.

- Materials such as vinyl, rubber flooring, cork, or carpet tiles are commonly used as they also help in reducing noise levels.
- Flooring should be durable and easy to maintain, considering the daily activities in the school.
- Different flooring textures can also be used carefully for sensory stimulation, but should not be too overwhelming. Overall, flooring should create a quiet, safe, and comfortable walking surface.

3.4 COLOUR PALETTE

- The color palette has a strong influence on mood, behavior, and concentration. Calm and neutral colors such as soft blues, greens, beiges, and pastels are preferred as they create a soothing environment
- . Bright and highly contrasting colors should be avoided as they may lead to overstimulation and distraction. However, controlled use of color can be applied for wayfinding, identifying different zones, or highlighting important areas.
- Consistency in color scheme throughout the space helps in creating a sense of familiarity and comfort. Matte finishes are preferred over glossy surfaces to reduce visual glare.

SENSORY SCHOOL FOR AUTISM

WHAT IS AUTISM

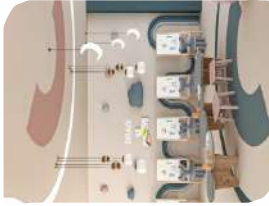
- Autism, or Autism Spectrum Disorder (ASD), is a developmental disorder that affects a person's social interaction, communication, and behavior.

CHARACTERISTICS OF AUTISM

- People with autism may show some of these traits (varies from person to person):
- Restricted interests: Deep focus on specific topics or hobbies.
- Sensory sensitivities: Over- or under-reacting to sounds, lights, textures, or touch.
- Need for routine: Comfort in predictable schedules; changes can cause stress.

CHALLENGES FACED BY AUTISM PEOPLE

- Autistic individuals may experience difficulties in:
- Education: Struggling with traditional classroom settings, group activities, or changes in routine.
- Communication: Expressing thoughts, understanding figurative language, or participating in discussions.
- Sensory Processing: Overwhelm from noisy, bright, or crowded environments.



- Light Grey: Acts as a neutral base, reducing visual clutter
- Soft Grey: Maintains simplicity and calmness
- Blush Pink: Adds subtle warmth without overstimulation
- Sage Green: Brings a natural and refreshing feel, improving comfort
- These colors create a peaceful, balanced, and nature-inspired space
- Muted Pink: Creates a warm, calming, and welcoming atmosphere
- Soft Blue: Promotes relaxation and reduces stress
- Beige / Neutral Tone: Adds balance and avoids visual overload
- Dark Blue (Accent): Provides contrast and depth without being too harsh
- This combination ensures a soothing and visually stable environment

01

SPECIAL STUDY



PROJECT INTRODUCTION

- An Autism School is a specialized learning environment designed for children with Autism Spectrum Disorder (ASD).
- These schools focus on creating structured, sensory-safe, and supportive spaces that help children learn, communicate, and develop daily living skills.
- The project aims to design an interior environment that reduces sensory overload, supports behavioral stability, and enhances independent functioning through thoughtful spatial planning, lighting, materials, and sensory zoning.

PURPOSE OF THE PROJECT

- Increasing number of children diagnosed with Autism Spectrum Disorder (ASD).
- Regular schools fail to support sensory, behavioral, and communication needs.
- Many schools lack therapeutic spaces, sensory rooms, and safe learning environments.
- Parents struggle to find institutions that focus on both education and behavioral development.
- A well-designed autism-friendly environment can significantly reduce anxiety, meltdowns, and stress for children.



- Vinyl flooring is a resilient flooring material with a thickness of 2-3 mm, finished with an anti-slip, matte surface.
- Carpet tiles are modular textile floor coverings with acoustic backing, providing excellent sound absorption and thermal comfort.
- Rubber flooring is proposed in high-activity and sensory spaces due to its shock-absorbing, anti-slip and sound-absorbing properties.
- Coral rock is a natural material formed from compressed coral and marine organisms over time. It has a porous texture and light color, making it suitable for interior and exterior applications.

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SENSORY SCHOOL FOR AUTISM

CONTEXTUAL CHALLENGES



- **Sensory Overload**
- Bright lights, loud sounds, and clutter overwhelm autistic children.
- Fix: Use soft lighting, acoustic control, and minimal design.
- **Confusing Circulation**
- Unclear pathways increase anxiety.
- Fix: Provide clear zoning, visual cues, and simple layouts.
- **Poor Acoustics**
- Echo and noise cause distress.
- Fix: Acoustic panels, soft flooring, quiet rooms.
- **No Sensory Regulation Spaces**
- Children have no area to calm down.
- Fix: Add sensory rooms, quiet corners, calming textures.
- **Safety Concerns**
- Sharp edges, open areas, and blind spots cause risk.
- Fix: Rounded edges, controlled access, better visibility.
- **Inadequate Outdoor Spaces**
- Regular playgrounds overstimulate or confuse.
- Fix: Sensory gardens, textured paths, quiet outdoor zones.
- **Accessibility Issues**
- Narrow spaces and poor signage reduce independence.
- Fix: Wide corridors, slip-resistant floors, pictorial signs.



- Lighting plays a crucial role in interior spaces by affecting visual comfort, mood, and functionality. A well-designed lighting system ensures proper illumination while avoiding discomfort such as glare or shadows.
- **Natural vs Artificial Lighting**
- Natural lighting (sunlight) creates a healthy and pleasant environment and helps reduce energy consumption.
- Artificial lighting (LED, tube lights) is used to maintain consistent illumination when natural light is insufficient.
- A balance of both improves comfort and efficiency.
- **Avoiding Glare and Flicker**
- Glare occurs when light is too bright or directly visible, causing eye strain.
- Flicker (unstable light) can cause discomfort and distraction.
- Proper placement of fixtures and use of high-quality lights help reduce these issues.



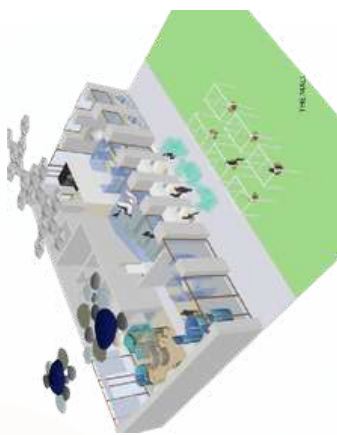
02

SPECIAL STUDY



DESIGN CONSIDERATIONS

- Structured, organized spaces for clarity and predictability
- Separate zones for learning, therapy, play, and relaxation
- Furniture that is safe, child-friendly, and low-height
- Storage spaces to reduce clutter and maintain order
- Adequate circulation space for children who may pace or need movement



SENSORY NEEDS

- Acoustic treatments to reduce noise and echoes
- Soft lighting, avoid glare or flickering lights
- Neutral color palettes to prevent overstimulation
- Textures and tactile elements to allow controlled sensory interaction
- Quiet or calm zones for children to regulate emotions



Use of Diffused Lighting

Diffused lighting spreads light evenly across the space. It reduces harsh shadows and creates a soft, comfortable environment.

Common methods include using false ceilings, panels, or frosted fixtures.

Color Temperature (Warm vs Cool Light)

Warm light (3000K–4000K): creates a cozy and relaxing atmosphere

Cool light (4000K–6000K): provides bright and focused illumination

Selection depends on the function of the space.

• Color Temperature (Warm vs Cool Light)

• Warm light (3000K–4000K): creates a cozy and relaxing atmosphere

• Cool light (4000K–6000K): provides bright and focused illumination

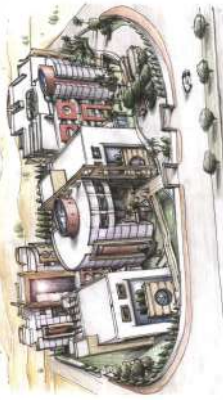
• Selection depends on the function of the space.



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SENSORY SCHOOL FOR AUTISM



MAGADA MUSTAFA

- Magda Mostafa is an Egyptian architect known for creating one of the first evidence-based design frameworks for autistic users. Her research focuses on how light, sound, space, and movement affect sensory regulation in children with autism.



- Magda Mustafa (Acoustic, Sensory Architecture for Autism) focuses on:
- ASPECTSS Design Index
- Acoustics – reduce noise, use sound-absorbing surfaces
- Spatial Sequencing – clear, step-by-step movement
- Escape Spaces – small rooms for calming down
- Compartmentalisation – specific zones for each activity
- Transition Spaces – buffers between functions
- Sensory Zoning – high-sensory vs low-sensory areas
- Safety – rounded edges, low-stimuli materials

03

SPECIAL STUDY



- Furniture Design
 - Use simple and functional furniture for easy usability
 - Prefer rounded edges to ensure safety
 - Choose lightweight and movable furniture for flexibility
 - Maintain comfortable seating height and proportions
 - Use soft finishes and non-reflective materials
- Layout Planning
 - Ensure clear circulation paths for smooth movement
 - Maintain adequate spacing between furniture
 - Avoid clutter to create a clean and organized space
 - Arrange furniture based on function and activity zones
 - Provide easy access to all areas



TYPES OF AUTISM HYPER AUTISM (OVER SENSITIVE).

- Features:
 - Small sounds feel very loud
 - Lights feel too bright
 - Touch may feel uncomfortable or painful
 - Strong smells or tastes may be overwhelming
 - Crowded spaces cause sensory overload
- Impact on behaviour:
 - Covering ears
 - Avoiding touch
 - Getting stressed in noisy / bright places
 - Preferring dim light, quiet rooms
 - Meltdowns due to too much stimulation
- Design Needs:
 - Soft lighting
 - Acoustic treatment
 - Neutral colours
 - Calm, low-stimuli spaces
 - Escape / quiet rooms

HYPO AUTISM (UNDER SENSITIVE).

- Features:
 - May not react to loud sounds
 - High pain tolerance
 - Seek strong touch or pressure
 - Might not sense temperature changes properly
- Enjoy spinning, jumping, bumping into objects
- Impact on behaviour:
 - Sensory-seeking actions
 - Touching everything
 - Enjoying bright lights
 - Craving movement (rocking, swinging)
- Design Needs:
 - Sensory rooms
 - Tactile materials
 - Deep-pressure furniture
 - Safe spaces for movement
 - Strong colour and texture variation

SOCIAL CHALLENGES

- Difficulty in communication (verbal + non-verbal)
- Trouble in understanding social cues like facial expressions
- Limited eye contact and difficulty in forming friendships
- Preference for routine → difficulty handling change
- Sensory overload in public places



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4. CASE STUDIES

4.1 V-EXCEL EDUCATIONAL TRUST

4.1.1 INTRODUCTION

- V-Excel Educational Trust is located in Chennai, Tamil Nadu, India. It is a well-known institution providing education and therapy for children with autism and other developmental disabilities.

4.1.2 DESIGN CONCEPT

- The design is based on a child-centered and inclusive approach, focusing on creating a safe, structured, and supportive environment. It integrates education, therapy, and daily life activities within a comfortable setting.

4.1.3 SPATIAL ORGANIZATION

- The planning follows a zoned and functional layout:
- Separate areas for classrooms, therapy rooms, and vocational training
- Clear circulation paths for easy navigation
- Open spaces for interaction and activities
- This helps children move independently and understand spaces easily.

4.1.4 MATERIAL PALETTE AND AESTHETIC

- Use of simple, durable, and locally available materials
- Soft and neutral color palette
- Minimal visual clutter to avoid distraction
- The aesthetic is functional, calm, and user-friendly.



Fig .4.1.4.a

4.1.5 LIGHTING DESIGN

- Maximum use of natural daylight
- Artificial lighting is soft and evenly distributed
- Avoidance of glare and flickering lights
- This supports comfort and reduces sensory stress.

• 4.1.6 FUNCTIONAL LAYOUT

- Spaces designed for learning, therapy, and skill development
- Flexible classrooms for different teaching methods
- Safe and accessible circulation
- The layout ensures smooth functioning and user safety.

• 4.1.7 CULTURAL INTEGRATION

- The design reflects local culture and community needs:
- Use of familiar materials and spatial arrangements
- Spaces encouraging group activities and social interaction
- Adaptation to local climate conditions

• 4.1.8 CONCLUSION

- V-Excel Educational Trust provides a balanced and inclusive environment, supporting education, therapy, and social development in a culturally responsive context.

- **4.2 UMMEED CHILD DEVELOPMENT CENTER**

- **4.2.1 INTRODUCTION**



- Ummeed Child Development Center is located in Mumbai, Maharashtra, India. It is a multidisciplinary centre providing therapy and developmental support for children with autism and other disabilities.

- **4.2.2 DESIGN CONCEPT**

- The design follows a family-centered and therapeutic approach, focusing on creating a welcoming and non-institutional environment. It supports both children and their families.

- **4.2.3 SPATIAL ORGANIZATION**

- Clear zoning of therapy, consultation, and waiting areas
- Organized circulation for easy movement
- Private and semi-private spaces for different activities
- This ensures comfort, privacy, and efficiency.

- **4.2.4 MATERIAL PALETTE AND AESTHETIC**

- Warm and neutral color tones
- Use of safe and child-friendly materials
- Soft textures and finishes
- The aesthetic is homely, calm, and comforting.

- **4.2.5 LIGHTING DESIGN**

- Good use of natural lighting
- Soft artificial lighting with controlled intensity
- Avoidance of harsh glare
- This creates a relaxed and stress-free environment.



Fig 4.2.5.a

• **4.2.6 FUNCTIONAL LAYOUT**

- Therapy rooms designed for specific needs
- Consultation spaces for parents and professionals
- Waiting and interaction areas
- Easy accessibility throughout the centre
- The layout supports both therapy and family involvement.

• **4.2.7 CULTURAL INTEGRATION**

- The centre reflects urban Indian context and family values:
- Spaces for parent interaction and counseling
- Familiar and comfortable environment
- Adaptation to compact urban settings

• **4.2.8 CONCLUSION**

- Ummeed Child Development Center creates a supportive and family-friendly environment, promoting therapy, interaction, and holistic development.

- **4.3 ANANDAM LEARNING CENTER**



- **4.3.1 INTRODUCTION**

- Anandam Learning Center is located in Ambattur, Chennai, India. It is a centre focused on providing special education and developmental support for children with autism.

- **4.3.2 DESIGN CONCEPT**

- The design focuses on a holistic and child-friendly approach, creating a safe, engaging, and flexible environment for learning and development.

- **4.3.3 SPATIAL ORGANIZATION**

- Zoning of classrooms, therapy, and play areas
- Use of open and semi-open spaces
- Clear and simple circulation paths
- Integration of indoor and outdoor activities
- This supports both structured learning and free play.

- **4.3.4 MATERIAL PALETTE AND AESTHETIC**

- Use of natural and locally sourced materials
- Soft, warm, and inviting color palette
- Safe and durable finishes
- The aesthetic is playful, comfortable, and welcoming.



Fig 4.3.4.a

- **4.3.6 LIGHTING DESIGN**

- Emphasis on natural daylight and ventilation
- Soft artificial lighting for indoor comfort
- Balanced illumination across spaces
- This enhances both comfort and usability.

- **4.3.6 FUNCTIONAL LAYOUT**

- Multi-functional classrooms
- Spaces for therapy, play, and interaction
- Safe and accessible movement
- Flexible layout for different activities
- The layout supports learning, therapy, and social interaction.

- **4.3.7 CULTURAL INTEGRATION**

- The design reflects local lifestyle and cultural context:
- Use of traditional materials and simple forms
- Inclusion of outdoor spaces
- Encouragement of group activities and community interaction

- **4.3.8 CONCLUSION**

- Anandam Learning Center provides a flexible, child-friendly, and culturally responsive environment, supporting overall growth and development of children with autism.



Fig 4.3.8.a

4.4 NEW STRUAN SCHOOL



4.4.1 INTRODUCTION

- New Struan School is located in Scotland, United Kingdom. It is a purpose-built school designed specifically for children with autism, focusing on providing a safe, structured, and supportive learning environment that addresses sensory, behavioral, and educational needs.

4.4.2 DESIGN CONCEPT

- The school is based on a low-arousal design concept, which aims to reduce sensory overload. Children with autism are sensitive to noise, light, and visual clutter, so the design minimizes these elements. The concept focuses on creating a calm, predictable, and controlled environment that helps improve concentration, behavior, and emotional stability.

4.4.3 SPATIAL ORGANIZATION

- The spatial planning follows a clear and linear organization, making navigation simple and stress-free:
- Classrooms are arranged in a sequence along corridors .Distinct zoning for learning, therapy, and recreational spaces

4.4.4 MATERIAL PALETTE AND AESTHETIC

- The material selection is focused on sensory comfort and safety: Use of soft, neutral, and pastel colors to avoid overstimulation
- Matte and non-reflective finishes to reduce glare .Acoustic materials like sound-absorbing panels to control noise



Fig 4.4.4.a

- **4.4.5 LIGHTING DESIGN**

- Lighting is carefully designed to support sensory needs:
- Maximum use of natural daylight through windows
- Artificial lighting is soft, diffused, and flicker-free
- Avoidance of harsh brightness and glare
- Consistent lighting levels to maintain comfort
- This helps in reducing anxiety and improving focus in children.

- **4.4.6 FUNCTIONAL LAYOUT**

- The layout is designed for efficiency and ease of use:
- Clear separation between quiet and active zones
- Classrooms designed for both group and individual learning
- Accessible circulation for smooth movement
- Safe and secure spaces for children
- The functional planning supports both learning activities and behavioral management.

- **4.4.7 CULTURAL INTEGRATION**

- The design follows a universal and user-centered approach rather than focusing on specific cultural elements. The emphasis is on creating a universally comfortable environment suitable for all children with autism.

- **4.4.8 CONCLUSION**

- New Struan School successfully demonstrates how a low-arousal, structured, and sensory-sensitive design can improve learning, behavior, and overall well-being of children with autism.



Fig 4.4.8.a

4.5 THE PUZZLE CENTRE



4.5.1 INTRODUCTION

- The Puzzle Centre is located in the United Kingdom.
- It is a specialized centre combining education, therapy, and research, focusing mainly on early intervention for children with autism.

4.5.2 DESIGN CONCEPT

- The design is based on the TEACCH (Treatment and Education of Autistic and Communication-Handicapped Children) approach. It emphasizes structured teaching, visual clarity, and routine-based learning. The aim is to help children understand activities independently through environmental cues.

4.5.3 SPATIAL ORGANIZATION

- The centre uses a highly structured spatial system:
- Clearly defined zones for learning, therapy, and play
- Use of partitions and furniture to create visual boundaries
- Flexible classroom layouts for different activities
- Organized circulation with minimal confusion
- This helps children easily understand space and function.

4.5.4 MATERIAL PALETTE AND AESTHETIC

- The material palette supports clarity and functionality:
- Neutral base colors with accent colors for guidance and zoning
- Durable and safe materials for frequent use
- Clean and clutter-free interiors
- Use of textures for sensory engagement
- The aesthetic is organized, functional, and visually structured.

4.5.5 LIGHTING DESIGN

Lighting is designed to be balanced and controlled:

Good use of natural lighting

Artificial lighting is evenly distributed

Avoidance of glare and high contrast

Comfortable brightness levels for visual clarity

This enhances focus and reduces sensory discomfort.

4.5.6 FUNCTIONAL LAYOUT

The layout supports independent and guided learning:

Clearly defined workstations

Spaces for both individual and group activities

Logical flow between different activity zones

Multi-functional classrooms

This improves usability and learning efficiency.

4.5.7 CULTURAL INTEGRATION

The design is research-oriented and functional, with limited cultural influence. The focus is on structured teaching methods rather than cultural expression.

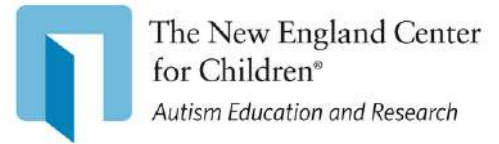
4.5.8 CONCLUSION

The Puzzle Centre effectively uses structured design principles and visual organization to support independent learning and early development in children with autism.



Fig 4.5.8 a

4.6 NEW ENGLAND CENTER FOR CHILDREN



4.6.1 INTRODUCTION

- New England Center for Children is located in the United States. It is one of the leading autism centres offering education, therapy, and residential programs for children.

4.6.2 DESIGN CONCEPT

- The design focuses on behavioral development, independence, and life skills. It combines structured educational spaces with a homely and comfortable environment, helping children adapt to real-life situations.

4.6.3 SPATIAL ORGANIZATION

- The centre follows a campus-style layout:
- Separate buildings for academic, residential, and therapy functions
- Clear zoning for different activities
- Inclusion of outdoor spaces for recreation and relaxation
- Safe and secure movement across the campus
- This organization supports both learning and daily living activities.

4.6.4 MATERIAL PALETTE AND AESTHETIC

- The material palette is designed to create a warm and welcoming environment:
- Use of warm, neutral colors
- Natural materials like wood
- Soft textures and finishes
- Comfortable and homely interiors
- The aesthetic reduces the feeling of an institutional space.



Fig 4.6.4.a

4.6.5 LIGHTING DESIGN

- Lighting enhances comfort and functionality:
- Extensive use of natural light
- Soft artificial lighting with controlled intensity
- Avoidance of harsh glare
- Warm lighting tones for a homely feel
- This supports both emotional comfort and visual clarity.

4.6.6 FUNCTIONAL LAYOUT

- The layout is designed for daily living and learning integration:
- Combination of classrooms and residential spaces
- Areas for therapy, recreation, and social interaction
- Safe and user-friendly circulation
- Spaces that promote independence and routine
- This helps children develop life skills along with education.

4.6.7 CULTURAL INTEGRATION

- The design reflects a residential and community-based approach, promoting social interaction and inclusion. Cultural aspects are subtle and focus more on creating a familiar and comfortable environment.

4.6.8 CONCLUSION

- The New England Center for Children successfully integrates education, therapy, and living spaces, creating a balanced environment that promotes independence, learning, and overall development.

4.7 ACTION FOR AUTISM



4.7.1 INTRODUCTION

- Action for Autism is located in New Delhi, India. It is one of the pioneering organizations in India working for individuals with autism, providing education, therapy, and advocacy services.

4.7.2 DESIGN CONCEPT

- The design focuses on a therapeutic and inclusive environment that supports learning, social interaction, and behavioral development. It emphasizes structured spaces with flexibility, allowing adaptation based on individual needs.

4.7.3 SPATIAL ORGANIZATION

- The spatial layout is zoned and activity-based:
- Separate areas for classrooms, therapy, and vocational training
- Clear circulation paths for easy movement
- Open and semi-open spaces for interaction
- This organization supports both structured learning and social engagement.

4.7.4 MATERIAL PALETTE AND AESTHETIC

- Use of natural and locally available materials
- Soft and neutral color palette to avoid sensory overload
- Durable and low-maintenance finishes
- The aesthetic is simple, functional, and user-friendly.



Fig 4.7.4.a

4.7.5 LIGHTING DESIGN

- Maximum use of natural daylight
- Artificial lighting is soft and evenly distributed
- Avoidance of glare and harsh lighting
- This creates a comfortable and stress-free environment.

• 4.7.6 FUNCTIONAL LAYOUT

- Spaces designed for education, therapy, and skill development
- Flexible classrooms for different activities
- Accessible and safe movement throughout the building
- The layout supports daily routines and functional independence.

• 4.7.7 CULTURAL INTEGRATION

- The design reflects Indian social and cultural context, including:
- Community-based spaces
- Use of local materials and familiar environments
- Encouragement of group interaction and inclusion

• 4.7.8 CONCLUSION

- Action for Autism creates a balanced and inclusive environment, supporting education, therapy, and social development within a culturally relevant setting.

4.8 SBI FOUNDATION – AUTISM CENTER



4.8.1 INTRODUCTION

- SBI Foundation Autism Center is located in Mumbai, India. It is an initiative aimed at providing specialized care, therapy, and training for children with autism.

4.8.2 DESIGN CONCEPT

- The design is based on a sensory-sensitive and child-centric approach, focusing on safety, comfort, and therapy. It integrates structured learning with calming environments.

4.8.3 SPATIAL ORGANIZATION

- Clear zoning for therapy, learning, and recreational spaces
- Defined circulation paths for easy navigation
- Inclusion of quiet rooms and sensory spaces
- This helps in reducing confusion and supporting different activity needs.

4.8.4 MATERIAL PALETTE AND AESTHETIC

- Use of soft colors and soothing tones
- Non-toxic and safe materials
- Smooth finishes with minimal visual clutter
- The aesthetic is calm, clean, and therapeutic.



Fig 4.8.4.a

- **4.8.5 LIGHTING DESIGN**

- Combination of natural and artificial lighting
- Use of diffused lighting to avoid harsh shadows
- Controlled brightness levels for comfort
- Lighting supports sensory regulation and focus.

- **4.8.6 FUNCTIONAL LAYOUT**

- Therapy rooms designed for specific activities
- Flexible learning spaces
- Safe and child-friendly circulation
- Areas for both individual and group sessions
- The layout ensures efficient functioning and user comfort.

- **4.8.7 CULTURAL INTEGRATION**

- The centre incorporates local context and community needs, with:
- Familiar spatial arrangements
- Spaces encouraging family involvement
- Adaptation to urban Indian conditions

- **4.8.8 CONCLUSION**

- The SBI Foundation Autism Center provides a well-structured and sensory-friendly environment, supporting therapy, education, and overall development.



Fig 4.8.8.a

- **4.9 PRENA CENTER (AUTISM CENTRE)**



- **4.9.1 INTRODUCTION**

- Prena Center is an autism support centre in India (exact location varies by reference). It focuses on therapy, special education, and skill development for children with autism.

- **4.9.2 DESIGN CONCEPT**

- The design follows a holistic and therapeutic approach, combining learning, play, and sensory experiences. It aims to create a safe, engaging, and adaptable environment.

- **4.9.3 SPATIAL ORGANIZATION**

- Zoning of spaces into learning, therapy, and play areas
- Use of open and semi-open spaces
- Clear and simple circulation
- Integration of indoor and outdoor environments
- This supports both structured and free activities.

- **4.9.4 MATERIAL PALETTE AND AESTHETIC**

- Use of natural materials and textures
- Soft and warm color palette
- Safe, durable, and child-friendly finishes
- The aesthetic is welcoming, playful, and comfortable.



Fig 4.9.4.a

- **4.9.5 LIGHTING DESIGN**

- Emphasis on natural lighting and ventilation
- Soft artificial lighting to avoid sensory stress
- Balanced illumination across spaces
- Lighting enhances comfort and usability.

- **4.9.6 FUNCTIONAL LAYOUT**

- Multi-functional classrooms and therapy spaces
- Areas for sensory play and interaction
- Flexible layout for different activities
- Safe and accessible design
- This supports learning, therapy, and development.

- **4.9.7 CULTURAL INTEGRATION**

- The design reflects local lifestyle and cultural context:
- Inclusion of outdoor play areas
- Use of traditional materials and simple forms
- Spaces encouraging social interaction

- **4.9.8 CONCLUSION**

- Prena Center provides a holistic and flexible environment, supporting the overall growth and development of children with autism through design.

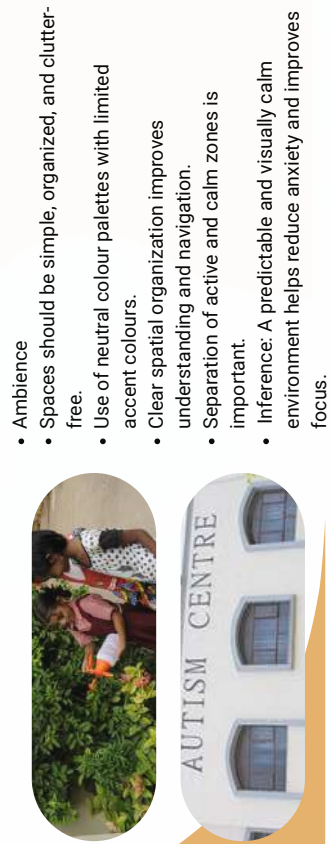
5.0 INFERENCE

SENSORY SCHOOL FOR AUTISM



INFERENCE

- Lighting
 - Lighting should be soft, diffused, and glare-free to avoid sensory discomfort.
 - Maximum use of natural light with proper shading control is preferred.
 - Artificial lighting should have warm colour temperature (3000K-4000K).
 - Avoid flickering lights and harsh contrasts.
- Zoning of lighting helps in defining different activity areas.
- Inference: Lighting must create a calm, comfortable, and non-overstimulating environment.
- 2. Materials
 - Materials should be non-toxic, safe, and durable.
 - Use of soft finishes like rubber flooring, cork, and fabric panels.
 - Edges and corners should be rounded to prevent injury.
 - Avoid shiny, reflective, or overly textured materials.
 - Inference: Materials should ensure safety, comfort, and minimal sensory disturbance.

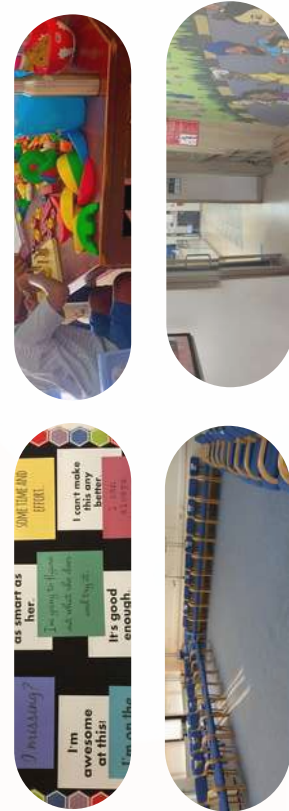


INFERENCE OF THE CASE STUDIES

- Ambience
 - Spaces should be simple, organized, and clutter-free.
 - Use of neutral colour palettes with limited accent colours.
 - Clear spatial organization improves understanding and navigation.
 - Separation of active and calm zones is important.
 - Inference: A predictable and visually calm environment helps reduce anxiety and improves focus.



- 4. Acoustics
 - Spaces should minimize external and internal noise.
 - Use of acoustic panels, soft furnishings, and sound-absorbing materials.
 - Noisy areas should be separated from quiet zones.
 - Avoid echo and reverberation.
 - Inference: Acoustic control is essential to prevent sensory overload and stress.
- 5. Concept & Form
 - Planning should follow simple, clear, and structured layouts.
 - Forms should avoid complexity; use clean lines and gentle curves.
 - Spatial sequencing should be logical and easy to understand.
 - Design should support routine and predictability.
 - Inference: Concept and form must promote clarity, ease of movement, and a sense of security.
- Wayfinding & Visual Cues
 - Use of colors, symbols, and signage for easy navigation.
 - Clear identity for each space.
 - Inference: Visual cues improve independence and orientation.



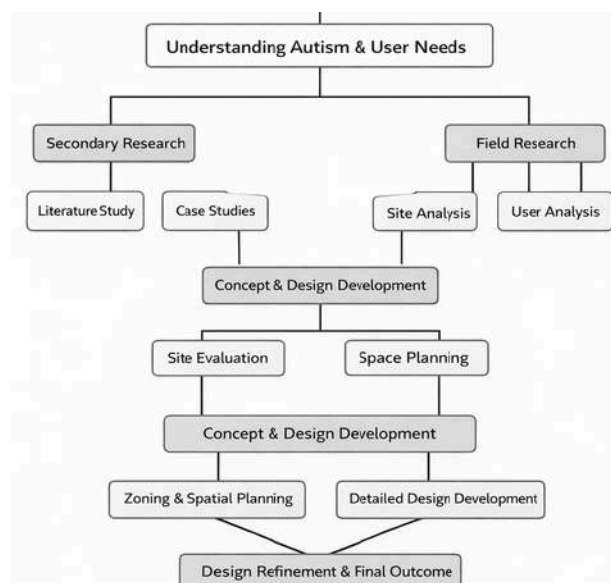
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Plate 5.0 .a Case Study Inference

5. METHODOLOGY

- The methodology for designing an autism school involves a systematic approach that combines research, analysis, and design development to create a functional and sensory-friendly environment.
- The process begins with understanding autism and user needs, where detailed study is conducted on the behavior, sensory sensitivities, and spatial requirements of children with autism.
- The next step involves literature study and data collection, where information is gathered from books, research papers, design guidelines, and standards related to autism-friendly environments.
- This helps in understanding concepts such as low-arousal design, sensory zoning, and structured learning spaces.
- Following this, case studies of existing autism schools and centers are analyzed to understand real-life applications of design principles, including spatial organization, material use, lighting, and circulation.
- Based on the collected data, site analysis is carried out to study the physical conditions of the proposed site, including location, climate, accessibility, and surroundings.
- This helps in making design decisions that are suitable for the context. After this, user analysis and space requirements are defined, identifying the types of spaces needed, their sizes, and their relationships with each other.



6 DESIGN DEVELOPMENT

6.1 USER ANALYSIS

- User analysis is a key step in designing an autism school, as the needs of users are very specific and sensitive. The primary users are children with autism, who may have different levels of sensory sensitivity, communication difficulties, and behavioral patterns.
- Some children may be hypersensitive to light, sound, and touch, while others may seek sensory stimulation. Therefore, spaces must be designed to either reduce or provide controlled sensory input based on individual needs.
- These children also require a structured and predictable environment, as sudden changes or confusing layouts can cause stress and anxiety.
- Secondary users include teachers, therapists, and caregivers, who need spaces that allow easy supervision, interaction, and flexibility in teaching methods.
- They require clear visibility across spaces, easy access to classrooms and therapy rooms, and safe movement throughout the building. Additionally, parents and visitors are also important users, especially in areas like reception, waiting zones, and counseling rooms.
- Their needs include comfort, privacy, and easy navigation.
- The analysis also considers behavioral patterns, such as the need for quiet spaces, retreat areas, and safe zones where children can calm down during sensory overload.
- Movement patterns are also important, as children benefit from simple, repetitive circulation paths. Overall, user analysis helps in creating a design that is safe, comfortable, functional, and supportive, improving both learning and daily activities.

6.2 PROXIMITY AND CIRCULATION

6.2.1 AUTISM SCHOOL

- Proximity and circulation play a very important role in the design of an autism school, as they directly affect how children understand, move through, and experience the space. In such environments, spaces must be arranged in a logical and structured manner so that children can easily relate one space to another without confusion.
- Proximity refers to the placement of spaces based on their function and relationship. Areas that are frequently used together, such as classrooms and therapy rooms, are located close to each other to reduce movement time and avoid unnecessary transitions.
- Public areas like reception and waiting zones are placed near the entrance for easy access and supervision, while more private areas like classrooms and therapy rooms are located deeper inside the layout to ensure safety and focus.
- Service areas such as toilets and storage are centrally located so they are easily accessible but visually unobtrusive.
-
- Circulation in an autism school is designed to be simple, clear, and predictable, helping children move independently and with confidence. Straight and linear circulation paths are preferred, as they are easy to understand and reduce confusion. In some cases, looped circulation is used to avoid dead ends, allowing continuous movement and minimizing frustration.
- The pathways are designed to be wide and clutter-free, ensuring safe and smooth movement, especially for group activities. Visual cues such as color coding, symbols, and signage are often used along circulation paths to support wayfinding and help children identify different zones.
- Transition spaces are also important, acting as buffers between different functional areas, allowing children to gradually adjust from one activity to another without sensory overload.
-

6.2.2 PROXIMITY AND BUBBLE DIAGRAM



1 1

Plate 6.2.2.a Proximity and Zoning

DEPARTMENT OF DESIGN
INSTITUTE OF SCIENCE & TECHNOLOGY, CHENNAI

The diagram illustrates a school building layout with various functional areas. The layout is organized into two main clusters of rooms, with a central legend defining the color-coding for different types of spaces.

Legend:

- Classes and administrative (Green)
- Therapy room (Blue)
- Cafe and life skills area (Orange)
- Clinical area (Purple)

Room Layout and Connections:

- Left Cluster:**
 - Reception** (Blue oval) is connected to **Waiting Area** (Light blue oval) and **Multipurpose Area** (Green oval).
 - Waiting Area** (Light blue oval) is connected to **Play Area** (Yellow oval) and **Diagnosis and Assessment Area** (Orange oval).
 - Play Area** (Yellow oval) is connected to **Waiting Area** (Light blue oval).
 - Diagnosis and Assessment Area** (Orange oval) is connected to **Waiting Area** (Light blue oval), **Office** (Grey rectangle), **Consultation Area** (Purple oval), and **Classroom 1** (Pink oval).
 - Office** (Grey rectangle) is connected to **Diagnosis and Assessment Area** (Orange oval) and **Multipurpose Area** (Green oval).
 - Multipurpose Area** (Green oval) is connected to **Reception** (Blue oval) and **Diagnosis and Assessment Area** (Orange oval).
 - Consultation Area** (Purple oval) is connected to **Diagnosis and Assessment Area** (Orange oval) and **Classroom 2** (Pink oval).
 - Classroom 1** (Pink oval) is connected to **Diagnosis and Assessment Area** (Orange oval) and **Classroom 2** (Pink oval).
 - Classroom 2** (Pink oval) is connected to **Diagnosis and Assessment Area** (Orange oval) and **Classroom 3** (Pink oval).
 - Classroom 3** (Pink oval) is connected to **Classroom 2** (Pink oval).
- Right Cluster:**
 - Vocational Training Area** (Blue oval) is the central hub, connected to **Art Therapy Room** (Purple oval), **Therapy Room** (Blue oval), **Sensory Room** (Blue oval), **Cookery** (Yellow oval), **Cafeteria** (Orange oval), **Outdoor Garden** (Green oval), and **Cafeteria** (Yellow oval).
 - Art Therapy Room** (Purple oval) is connected to **Vocational Training Area** (Blue oval).
 - Therapy Room** (Blue oval) is connected to **Vocational Training Area** (Blue oval) and **Sensory Room** (Blue oval).
 - Sensory Room** (Blue oval) is connected to **Vocational Training Area** (Blue oval) and **Cookery** (Yellow oval).
 - Cookery** (Yellow oval) is connected to **Vocational Training Area** (Blue oval) and **Cafeteria** (Orange oval).
 - Cafeteria** (Orange oval) is connected to **Vocational Training Area** (Blue oval) and **Cafeteria** (Yellow oval).
 - Outdoor Garden** (Green oval) is connected to **Vocational Training Area** (Blue oval).
 - Cafeteria** (Yellow oval) is connected to **Vocational Training Area** (Blue oval).

- Reception and Waiting areas are placed at the entrance for easy access and supervision.
- Therapy and multipurpose areas are centrally located to connect play, art, and sensory activities efficiently.
- Clinical areas like Diagnosis and Consultation are positioned near classrooms for convenient assessment and observation.
- Vocational and life skills areas (Cookery, Sky Juice, Outdoor Garden) are designed to promote independence and real-world learning.

- [illegible]

NOT NEAR NEARBY NEAR

10 BUBBLE DIAGRAM AND PROXIMITY

Name: V.KSHIRAJA

REG NO: 42102008

6.3 AREA STATEMENT

AREA STATEMENT

- The area statement is prepared based on the needs of autistic children for learning, therapy and daily life activities.
- Spaces are planned with clear zoning so that public, learning and therapy areas do not disturb each other.
- Classrooms, therapy rooms and activity areas are supported by outdoor spaces to create a calm and comfortable environment.
- The areas mentioned are approximate and will be adjusted during further design development.

AREA STATEMENT NEEDED IN DESIGN

- An area statement is needed to understand and organise the required spaces and their sizes based on the design concept and user needs.
- It helps the designer plan spaces according to activities, comfort and circulation. The area statement ensures that no important space is missed and that all spaces are correctly proportioned.
- It also helps in zoning, furniture planning and efficient use of the site.

PURPOSE

- Helps translate concept into actual spaces
- Helps decide size of each space
- Helps in zoning and circulation planning
- Helps in furniture layout and space planning
- Helps maintain sensory comfort and safety
- Helps estimate total built-up area



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SENSORY SCHOOL FOR AUTISM

S. No	Space	User	Approx. Area (sqm)	Function
1	Reception	Parents, Visitors	40	Entry and enquiry
2	Play Area	Children	60	Physical play activities
3	Multipurpose Area	Children, Staff	100	Group activities and events
4	Diagnosis & Assessment Area	Child, Therapist	30	Evaluation and assessment
5	Waiting Area	Parents	40	Waiting and transition
6	Consultation Area	Parents, Therapist	25	Counselling and guidance
7	Classroom 1	Students, Teacher	50	Academic learning
8	Office	Staff	30	Administration
9	Classroom 2	Students, Teacher	50	Academic learning
10	Classroom 3	Students, Teacher	50	Academic learning
11	Sensory Room	Students	25	Sensory regulation
12	Art Therapy Room	Students	35	Creative therapy
13	Outdoor Garden	Students	150	Sensory and therapeutic garden
14	Cookery	Students, Trainer	40	Life-skill training
15	Cafeteria	Students, Staff	80	Dining
16	Vocational Training Area	Students	60	Skill development

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AREA STATEMENT

06

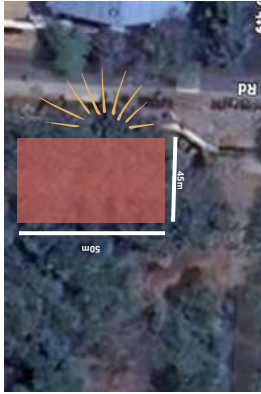
Plate 6.3.a Area Statement

6.4 SITE JUSTIFICATION

SENSORY SCHOOL FOR AUTISM

SITE INTRODUCTION

- Total site area: 2600 sqm (approx)
- Built up area: 2250 sqm
- Length: 50 m
- Breadth: 45 m
- Address: Context Adjacent to Anna Nagar 3rd Main Rd / Tower Park
- Location: Anna Nagar



OVERVIEW

- The site is easily accessible through major and secondary roads, reducing travel stress for autistic children.
- The surrounding environment is calm and low-noise, suitable for sensory-sensitive users.
- Proximity to hospitals and therapy centers ensures quick medical and emergency support.
- The site is located within a safe residential/institutional neighborhood.
- Adequate plot size allows proper zoning of classrooms, therapy spaces, and sensory areas.
- Regular site shape supports simple planning and clear circulation.
- Availability of natural light and ventilation enhances sensory comfort.
- Presence of green surroundings supports biophilic and calming design.
- Controlled entry and exit points improve safety and security for children.
- The site offers scope for future expansion of special education facilities.



04

SITE JUSTIFICATION

SITE LOCATION AND JUSTIFICATION

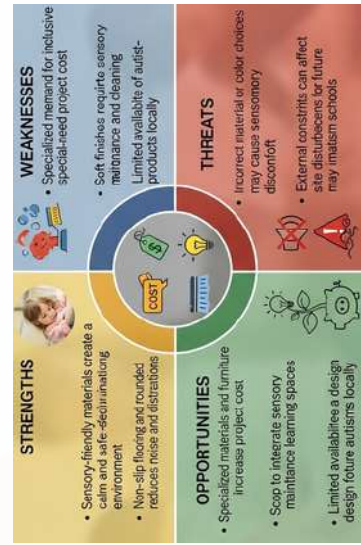
- The proposed site is located in an urban residential area with good connectivity to main roads and public transport, making it easily accessible for parents, staff, and therapists.
- The surrounding context includes schools, hospitals, parks, and residential neighborhoods, which supports the functional needs of an autism learning center. The site is relatively calm and away from heavy traffic and industrial activities, helping to reduce noise and sensory disturbance for autistic children.
- The flat terrain and regular site shape allow barrier-free movement and flexible planning.
- Adequate natural light, ventilation, and scope for outdoor spaces make the site suitable for creating a safe, inclusive, and sensory-friendly learning environment, justifying the selection of this site for the proposed autism learning center.



TARGET USERS

- **Children with Autism Spectrum Disorder (ASD)**
 - Primary users of the facility, including children with varying levels of sensory, behavioral, and learning needs.
- **Therapists and Special Educators**
 - Occupational therapists, speech therapists, psychologists, and special educators who conduct therapy, training, and assessment.
- **Parents and Caregivers**
 - Parents and guardians who accompany children for therapy sessions, consultations, and review meetings.
- **Administrative and Support Staff**
 - Office staff, coordinators, and maintenance personnel responsible for daily operations and management.
- **Visiting Professionals**
 - Medical practitioners, counselors, and trainers who visit periodically for assessments and workshops.

SWOT ANALYSIS



REG NO: 42102008

Name: V.KSHIRAJA

Plate 6.4.a Site Justification

6.5 CONCEPT DEVELOPMENT

- The concept development of the autism school is based on a deep understanding of the behavioral, sensory, and cognitive needs of children with autism. Research and case studies indicate that such children are highly sensitive to environmental stimuli such as light, sound, color, and spatial complexity.
- The design concept is developed around the idea of “structured simplicity”, where spaces are organized in a clear and logical manner to avoid confusion and disorientation. Zoning is carefully planned by dividing the school into public (entrance, reception), semi-private (classrooms, therapy), and private zones (sensory rooms, quiet spaces).
- This hierarchy ensures safety, privacy, and smooth functioning. Spaces that require frequent interaction, such as classrooms and therapy rooms, are placed in close proximity, while noisy areas like play zones and cafeterias are separated from quiet zones to maintain a balanced sensory environment.
- The concept also emphasizes clear and predictable circulation, which plays a major role in reducing stress and promoting independence. Linear and looped pathways are used to ensure easy navigation, while visual cues such as color coding and signage support wayfinding.
- Transition spaces are introduced between different zones to act as buffers, allowing children to gradually adapt from one activity to another without sudden sensory changes.
- In addition, the design integrates nature and outdoor spaces as part of the learning environment.
- Gardens, play areas, and semi-open spaces provide opportunities for relaxation, sensory engagement, and social interaction. These spaces help in improving emotional well-being and offer a break from structured indoor activities.
- Overall, the concept development aims to create a safe, inclusive, and therapeutic environment that supports not only academic learning but also behavioral development, independence, and social interaction.

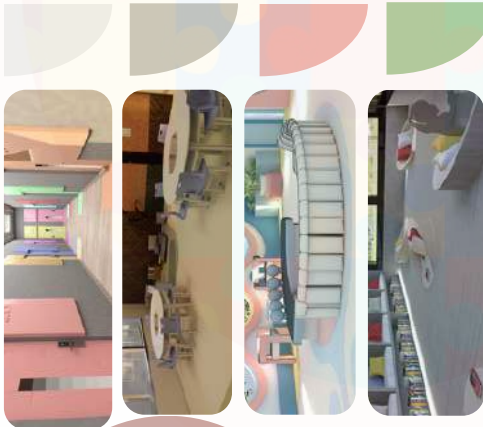
SENSORY SCHOOL FOR AUTISM

ABOUT

- The concept for the Autism School is based on creating a sensory-responsive, safe and predictable environment that supports the emotional, cognitive and sensory needs of children with Autism Spectrum Disorder (ASD).
- The design focuses on reducing sensory overload while providing controlled sensory engagement through space planning, materials, lighting and furniture.

- Spaces are designed to follow clear zoning and smooth transitions, helping children move from one activity to another without stress or confusion.
- Each space responds differently to hyper-sensory and hypo-sensory needs, allowing children to feel calm, secure and supported.

- The concept draws inspiration from Magda Mostafa's Sensory Design Theory, emphasizing low stimulus interiors, visual clarity, spatial hierarchy and sensory balance to enhance learning, therapy and social interaction



- **Sensory Zoning**
- High-stimulus spaces (play, therapy) are separated from low-stimulus spaces (classrooms, calming rooms).
- Quiet zones are placed away from noisy or active areas.
- **Predictability & Routine**
- Simple circulation paths
- Clear entry points and wayfinding
- Repetition in spatial layout to reduce anxiety

- **Controlled Sensory Engagement**
- Use of soft textures, muted colours and indirect lighting
- Sensory rooms provided for regulated stimulation
- Outdoor interaction through courtyards and gardens
- **Safety & Comfort**
- Rounded furniture edges
- Soft flooring and non-slip materials
- Visual supervision without overstimulation

07

CONCEPT AND THEME



MINIMALIST, BIOPHILIC & SENSORY RESPONSIVE DESIGN

- The theme focuses on creating a calm, predictable and nurturing learning environment for autistic children by combining minimalist design principles, biophilic connections to nature, and sensory-responsive spatial planning

- **Minimalist Design:**
 - Simple and clutter-free spaces
 - Clear spatial organization and layouts
 - Neutral and muted colour tones
- **Biophilic Design:**
 - Strong connection between indoor and outdoor spaces
 - Use of natural light and ventilation
 - Integration of courtyards, gardens and green views
 - Use of natural materials such as wood and stone
 - Visual access to nature from learning and therapy spaces
- **Sensory-Responsive Design:**
 - Controlled lighting to avoid glare and harsh contrasts
 - Acoustic comfort through sound-absorbing materials
 - Soft textures and non-irritating finishes
 - Clear zoning of noisy and quiet spaces
 - Dedicated sensory rooms for regulation

Concept Rational:

- The concept of Minimalist, Biophilic and Sensory-Responsive design was selected to reduce sensory overload and create a calm, predictable learning environment for autistic children. Minimalism minimizes visual distractions, biophilic design promotes comfort through nature, and sensory-responsive planning ensures controlled light, sound and textures to support learning and well-being.



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6.6 MOOD BOARD

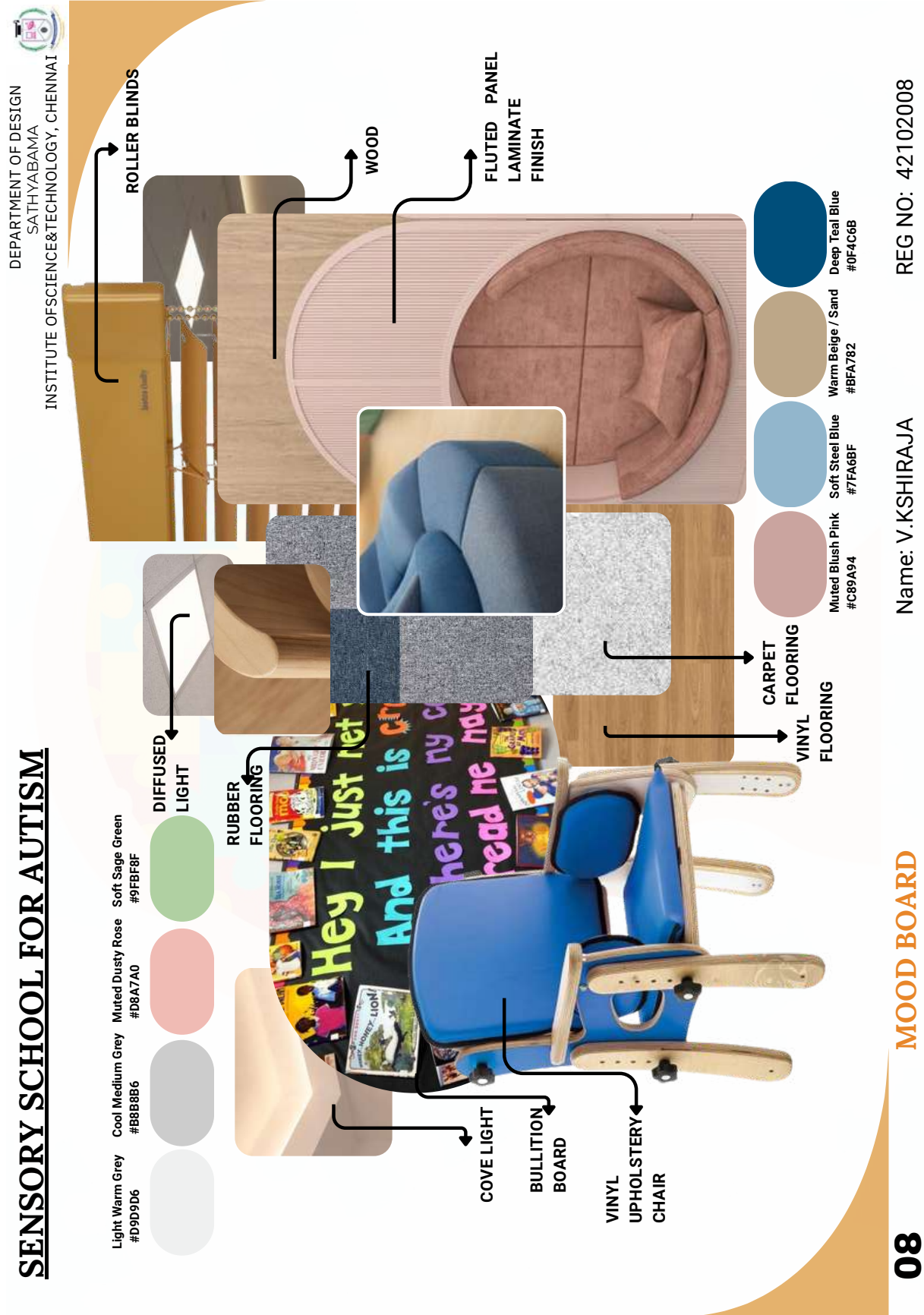
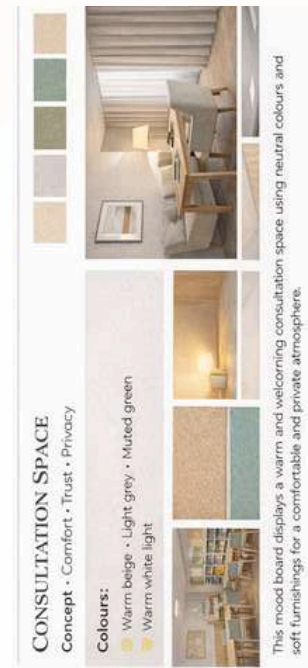
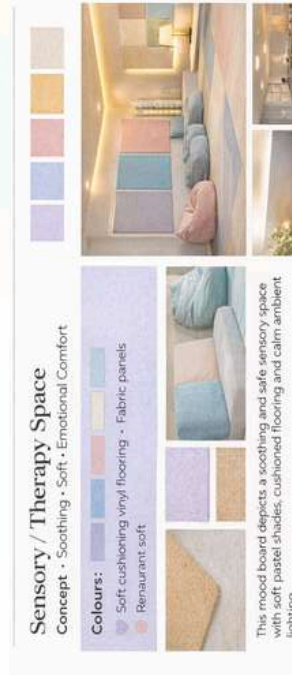
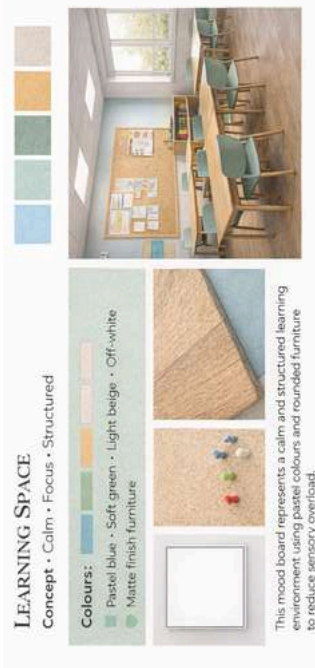


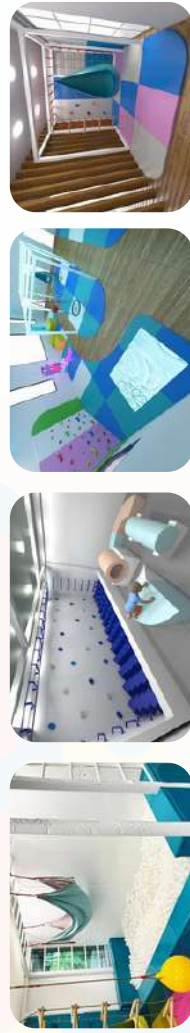
Plate 6.6.A Mood Board

SENSORY SCHOOL FOR AUTISM

6.6 MOOD BOARD



- The learning space is designed to be calm and structured, helping children focus without distractions. It uses pastel colours and a simple layout with clearly defined zones
- Furniture is safe and ergonomic, and lighting is soft and natural to create a comfortable environment that improves concentration and reduces sensory overload.



- The consultation space provides a private and comfortable setting for one-to-one interaction.
- It uses warm neutral colours, cozy seating like sofas or armchairs, and minimal elements to avoid distraction. Soft, warm lighting creates a relaxed atmosphere that helps build trust and supports effective communication.



- The therapy space is designed to be soothing and sensory-friendly, supporting relaxation and emotional regulation.
- It features soft pastel colours, cushioned flooring, and sensory elements like panels and bean bags.
- Gentle ambient lighting and a safe environment help reduce stress and improve sensory processing.

7 SCHEMATIC PLAN DEVELOPMENT

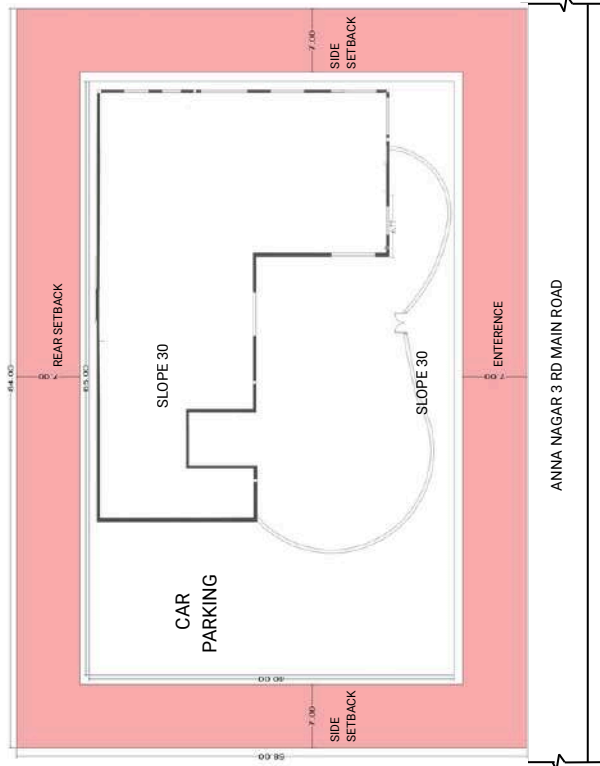
7.1.1 GROUND FLOOR PLAN

- The ground floor is designed to accommodate public and semi-public functions, ensuring easy accessibility and smooth entry for users.
- The reception is placed near the main entrance as the primary point of contact, along with the waiting area to provide comfort for parents and visitors.
- Adjacent to this, the consultation area and diagnosis and assessment area are located to allow quick and efficient interaction between professionals and users, while maintaining a level of privacy.
- The play area and multipurpose area are positioned close to the entrance zone but slightly separated from quiet spaces, as these are active zones with higher noise levels.
- The outdoor garden is connected to the play area, providing a natural extension for recreational and sensory activities.
- The cafeteria and cookery area are placed in a semi-accessible zone, ensuring ease of use while being slightly distanced from academic spaces to avoid disturbance.
- The vocational training area is also included on the ground floor, allowing easy access for all age groups and encouraging skill-based learning.
- Corridors are designed in a simple and linear manner, ensuring clear movement and easy navigation. The spatial arrangement follows a logical sequence, minimizing confusion and supporting independent movement for children with autism.

• **7.1.2 FIRST FLOOR PLAN**

- The first floor is planned as a more private and quieter zone, dedicated mainly to academic and therapeutic functions.
- Classroom 1, Classroom 2, and Classroom 3 are arranged in close proximity to ensure easy supervision and efficient movement between learning spaces. These classrooms are placed away from noisy areas to maintain a focused and calm learning environment.
- The sensory room and art therapy room are located near the classrooms, allowing quick access for therapeutic activities and providing a supportive environment for emotional and sensory regulation.
- These spaces are positioned in quieter corners to avoid disturbance.
- The office and staff area are placed strategically to provide visual control over circulation spaces and classrooms, ensuring safety and supervision. Corridors on this floor are designed to be wide, clutter-free, and clearly defined, supporting safe and independent movement.
- The overall layout of the first floor emphasizes structured zoning, minimal distraction, and controlled sensory experience, ensuring that children can learn and engage in activities comfortably.
- The schematic plan development ensures a clear division between active and quiet zones, with proper placement of public, semi-private, and private spaces.
- The design supports easy circulation, functional efficiency, and sensory comfort, creating a structured and user-friendly environment for children with autism.

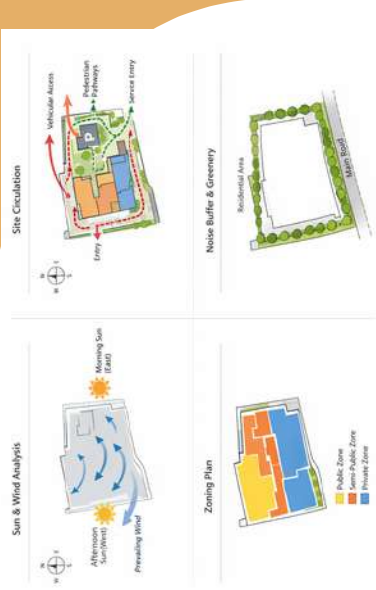
SENSORY SCHOOL FOR AUTISM



- The site is located within an urban residential zone, ensuring easy accessibility for children, parents, therapists, and staff.
- The surrounding context consists mainly of low- to mid-rise residential and institutional buildings, which helps maintain a relatively calm and controlled environment suitable for an autism school.
- The site has good connectivity to primary and secondary roads, allowing smooth vehicular and pedestrian access while minimizing heavy traffic disturbance.
- Ambient noise levels are moderate and can be further controlled through appropriate zoning and landscape buffers.
- The site features a largely flat terrain, which supports barrier-free design and safe movement for children with special needs.
- Adequate natural light and cross-ventilation are achievable due to open setbacks and surrounding building heights, contributing to a healthy indoor environment.
- The orientation of the site allows controlled daylight entry, reducing glare and visual discomfort.
- Overall, the site conditions support the creation of a safe, sensory-friendly, and inclusive learning environment, making it suitable for an autism learning and therapy center.

05

SITE PLAN



ABUTTING ROAD

- The site directly abuts Anna Nagar 3rd Main Road.
- The road provides primary access to the site.
- The road has moderate traffic and functions as an internal arterial road.
- Suitable for school entry, drop-off and pick-up zones.

ROAD WIDTH

- Anna Nagar 3rd Main Road: approx. 9–12 m wide
- Nearby cross streets (4th St, 6th St): approx. 6–7 m wide
- Adequate turning radius and vehicular access.

FIRE DETECTION

- Smoke detectors in classrooms, corridors, therapy rooms.
- Fire alarm system should:
- Have visual alerts (flashing lights)
- Avoid excessively loud sirens.
- Audible alarms can be combined with visual cues.

SITE ORIENTATION

- The site is oriented along a North–South axis.
- Road frontage faces approximately East.
- Open areas are towards North and West for controlled daylight.
- Orientation allows good daylight penetration and cross-ventilation.

FIRE SAFETY

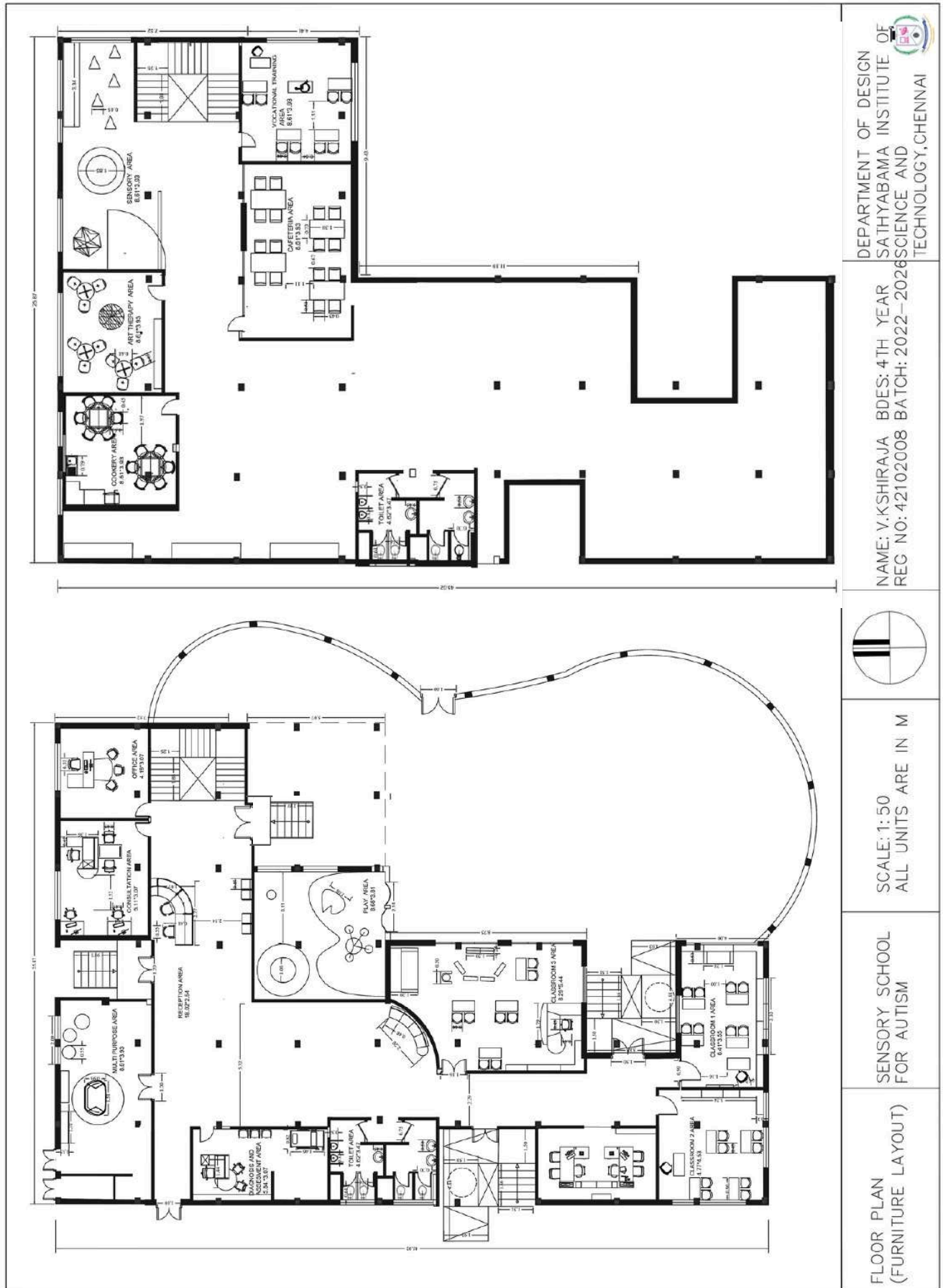
- Fire safety design in an autism school prioritizes clear access, simple evacuation routes, controlled alarms, fire-safe materials, and assisted evacuation, ensuring safety without causing sensory distress.



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Plate 7.2.a Site Plan



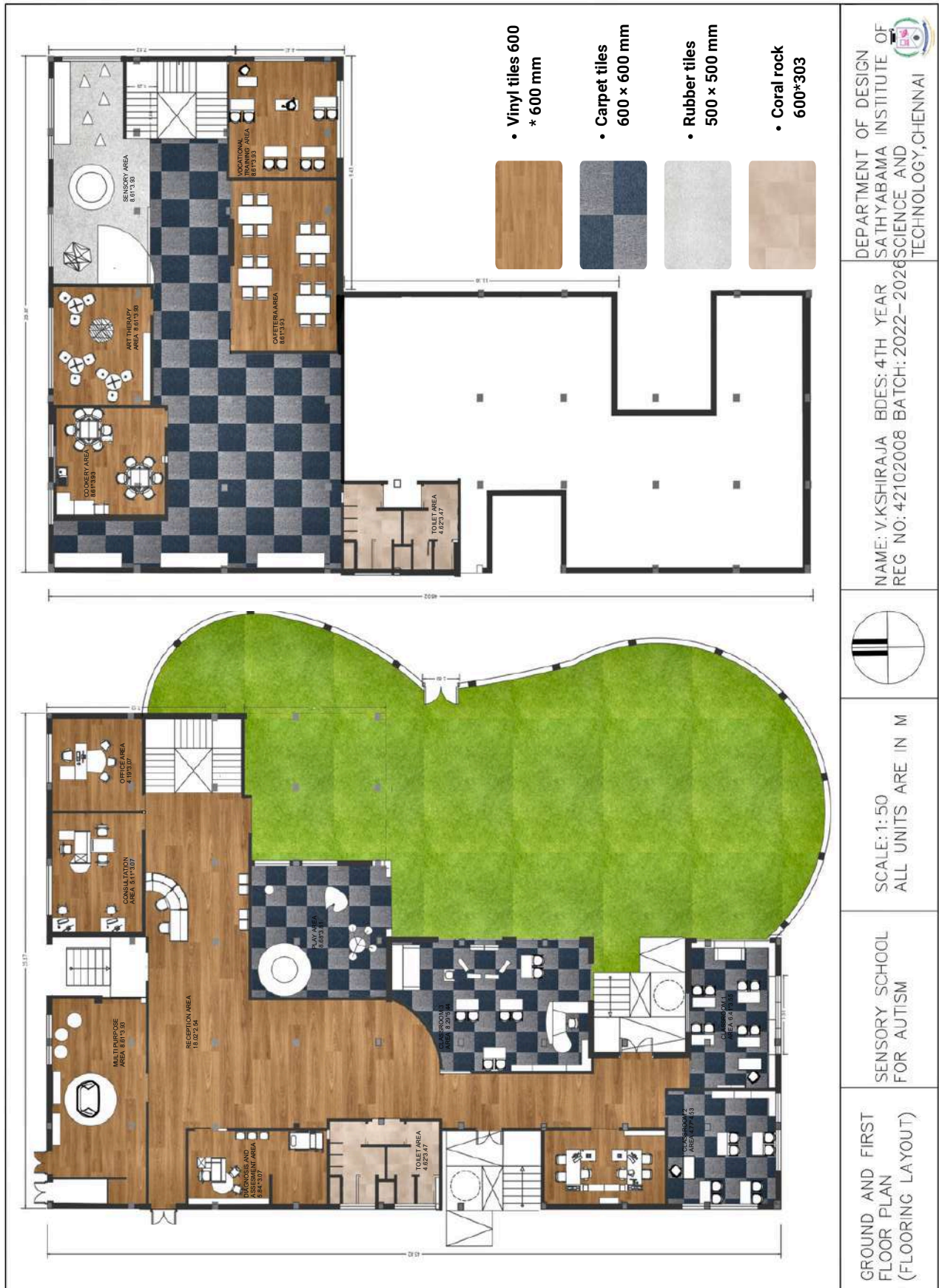
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Plate 7.2.b Furniture Layout



18

Plate 7.2.d Electrical Layout



16

Plate 7.3.a Flooring Layout



Plate 7.3.b Floor Render

8.DISCUSSION AND INFERENCE

• 8.1 CONCEPTUAL STAGE

- At the conceptual stage, the design approach is developed based on the understanding of autism and user needs derived from literature and case studies. It is inferred that children with autism require a structured, predictable, and low-arousal environment.
- The concept focuses on minimizing sensory overload by controlling light, sound, and visual distractions. Key ideas such as clear zoning, simple circulation, and calm spatial experience are established.
- The importance of creating a safe and comfortable environment that supports both learning and behavioral development is identified at this stage. This forms the foundation for all further design decisions.

• 8.2 SCHEMATIC STAGE

- During the schematic stage, the conceptual ideas are translated into spatial arrangements and zoning diagrams. Based on case study analysis, it is inferred that spaces must be organized into public, semi-private, and private zones.
- Classrooms are placed close to therapy areas for easy access, while noisy spaces like play areas and cafeterias are located away from quiet zones. Circulation is kept linear and simple to ensure easy navigation.
- Bubble diagrams and adjacency studies help in understanding the relationship between different spaces. This stage ensures that the layout is logical, functional, and user-friendly.

8.3 DESIGN DEVELOPMENT STAGE

- In the design development stage, the schematic layout is further refined by focusing on materials, lighting, furniture, and detailed spatial design. It is inferred that the use of soft color palettes, non-reflective materials, and acoustic treatments is essential to create a sensory-friendly environment.
- Lighting design avoids glare and uses diffused illumination to maintain comfort. Furniture is designed to be safe, ergonomic, and adaptable to different activities.
- Circulation spaces are widened and simplified to ensure safe movement. This stage integrates all design elements to create a cohesive and functional environment.

8.4 FINAL DESIGN STAGE

- The final design stage brings together all the developed elements into a complete and refined design solution. It is inferred that a successful autism school design must balance functionality, safety, sensory comfort, and aesthetics.
- The final layout ensures clear zoning, smooth circulation, and effective use of space. The design supports learning, therapy, and social interaction while maintaining a calm and structured environment.
- Overall, the final design reflects a holistic approach that enhances the well-being, independence, and development of children with autism.

9. FINAL DESIGN

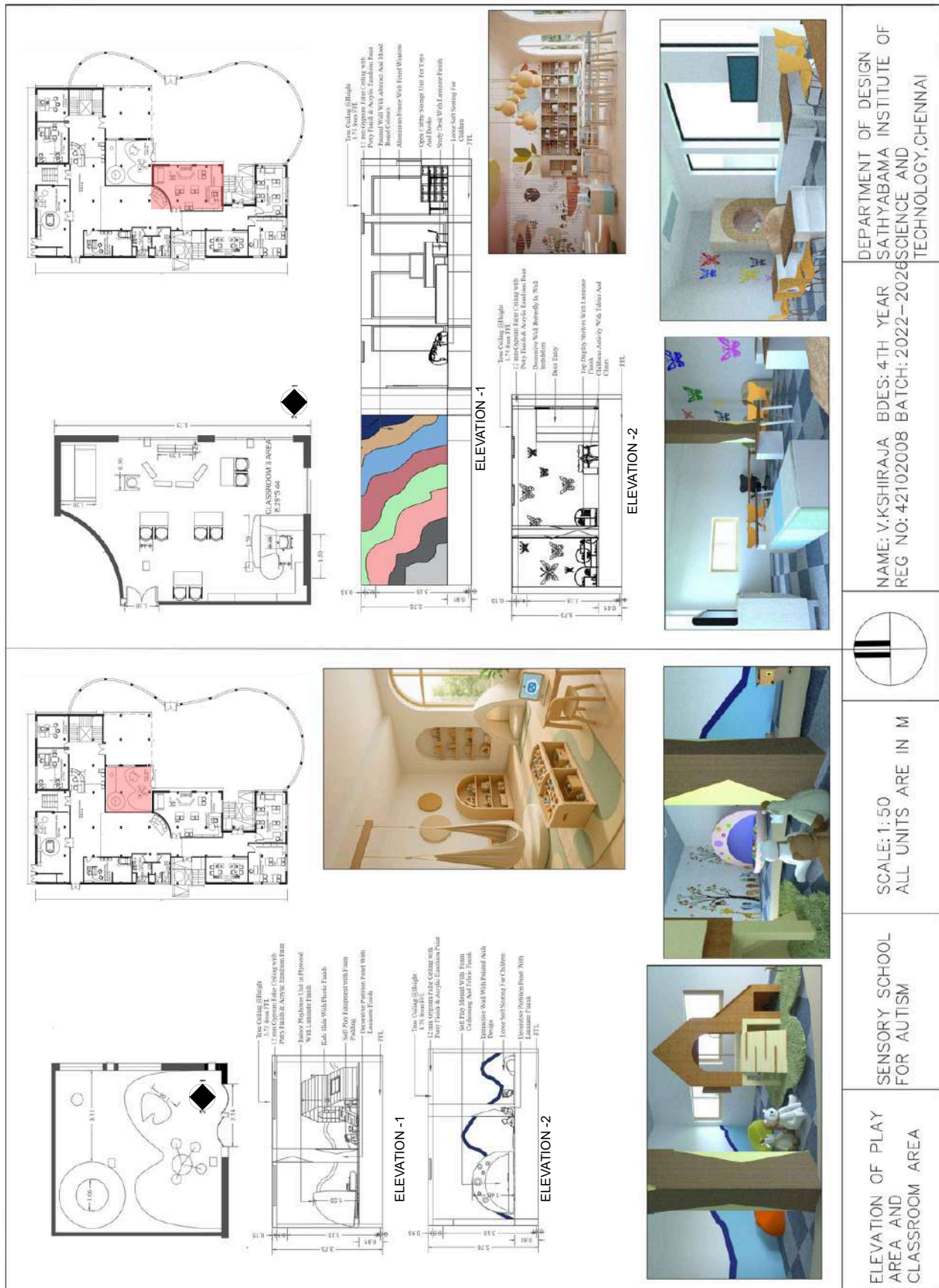


Plate 9.a Play Area, Classroom

		ELEVATION OF MULTIPURPOSE ROOM, CONSULTATION AND OFFICE ROOM	NAME: V.KSHIRAJA BDES: 4TH YEAR REG NO: 42102008 BATCH: 2022-2026	DEPARTMENT OF DESIGN SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI
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Plate 9.b Multipurpose Area ,Consultation Area, Office Area

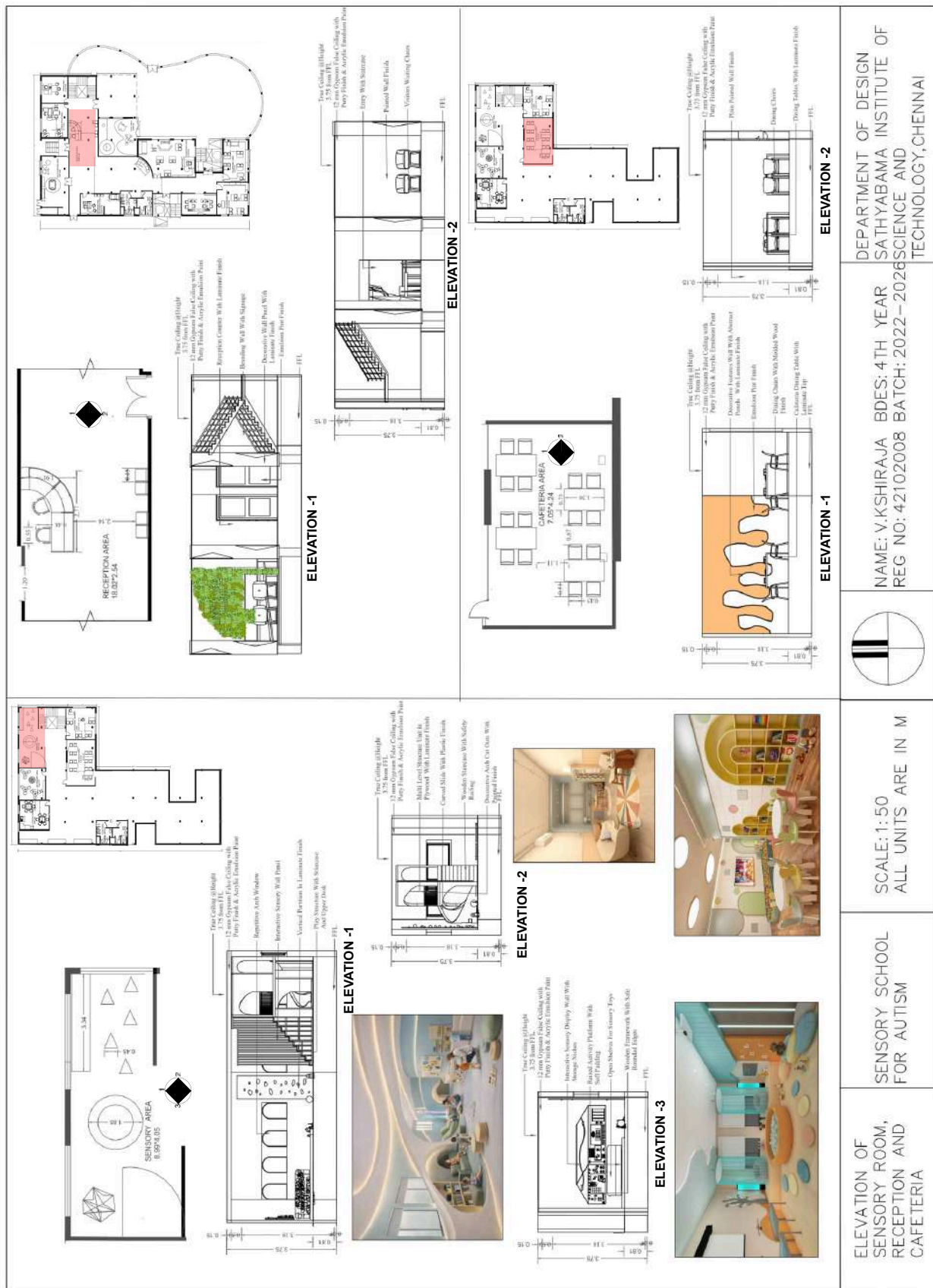


Plate 9.c. Sensory Area, Reception , Cafeteria

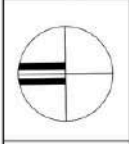
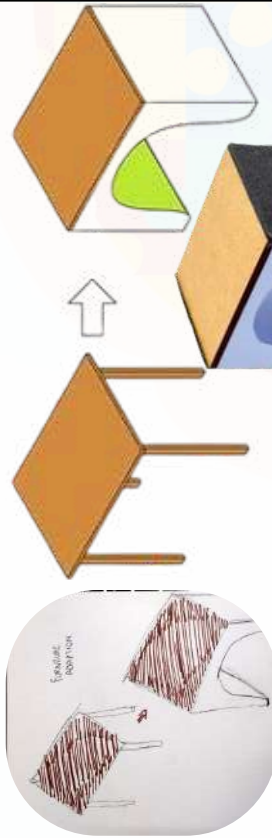
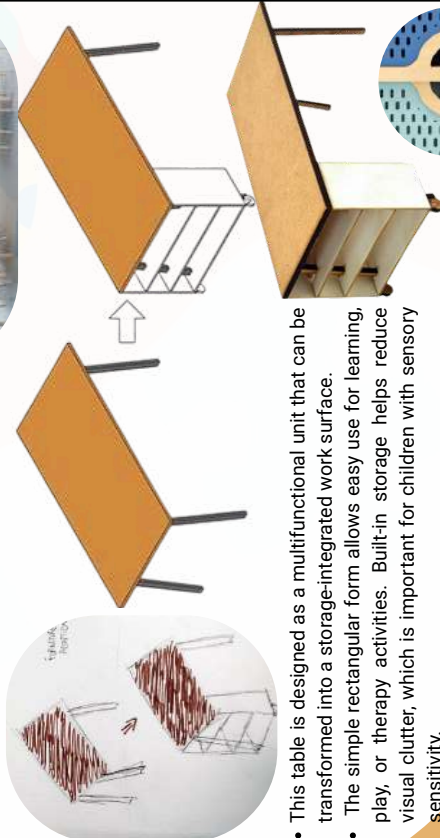
		RECEPTION VIEWS				
		CAFETERIA VIEWS				
		SENSORY AREA VIEWS				
VIEWS OF RECEPTION, SENSORY AND CAFETERIA		SENSORY SCHOOL FOR AUTISM	SCALE: 1:50 ALL UNITS ARE IN M		NAME: V.KSHIRAJA BDES: 4TH YEAR REG NO: 42102008 BATCH: 2022-2026	DEPARTMENT OF DESIGN SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI

Plate 9.d Views

SENSORY SCHOOL FOR AUTISM



- The board unit is designed to function both as a writing surface and a storage element, reducing the need for additional furniture.
- The inclined board angle supports comfortable writing and visual clarity. Integrated shelves allow teaching aids and materials to be stored neatly, promoting organization and structured learning.
- The minimal and clean design helps maintain a calm visual environment for autistic children.



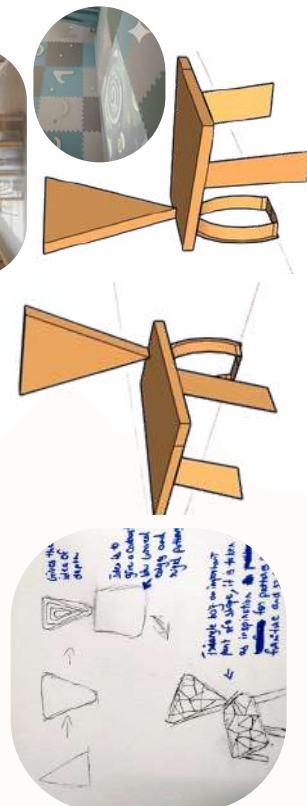
- This table is designed as a multifunctional unit that can be transformed into a storage-integrated work surface.
- The simple rectangular form allows easy use for learning, play, or therapy activities. Built-in storage helps reduce visual clutter, which is important for children with sensory sensitivity.
- The sturdy legs and clear leg space ensure accessibility and comfort, while the adaptable design supports flexible classroom arrangements.



13 FURNITURE DESIGN



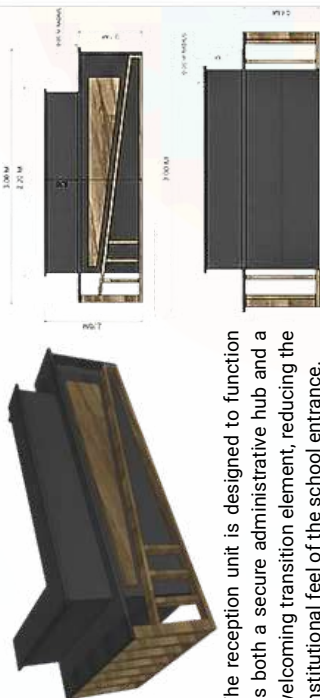
- The board unit is designed to function both as a writing surface and a storage element, reducing the need for additional furniture.
- The inclined board angle supports comfortable writing and visual clarity. Integrated shelves allow teaching aids and materials to be stored neatly, promoting organization and structured learning.
- The minimal and clean design helps maintain a calm visual environment for autistic children.



- The chair is designed with a stable wooden frame and soft, angular geometry to support proper posture while ensuring safety for children with autism.
- The wide base and curved supporting members provide balance and reduce the risk of tipping. Rounded edges and smooth finishes minimize injury and sensory discomfort.
- The use of warm-toned natural wood helps create a calming and non-intimidating seating experience, encouraging children to sit comfortably for longer durations.

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SENSORY SCHOOL FOR AUTISM



• The reception unit is designed to function as both a secure administrative hub and a welcoming transition element, reducing the institutional feel of the school entrance.

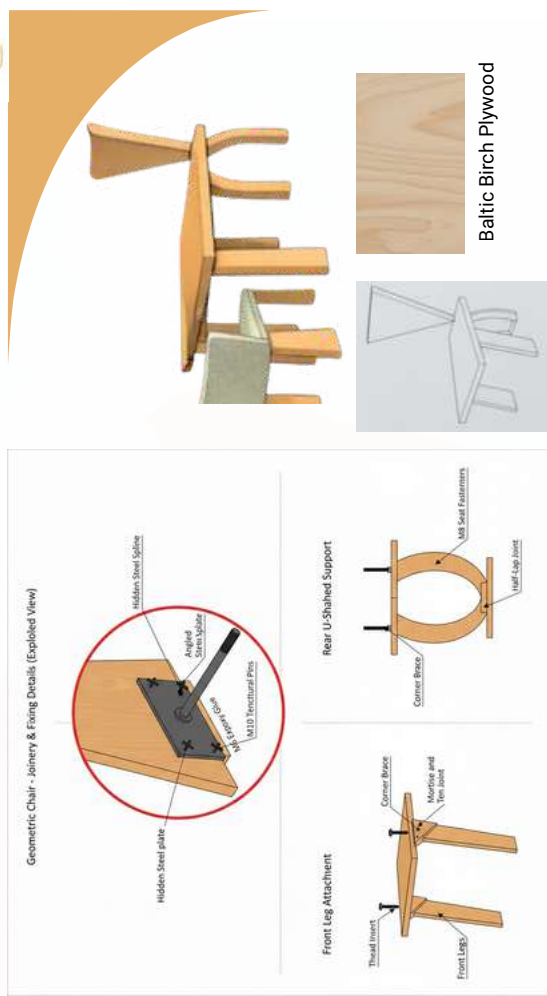
• **Dual-Height Architecture:** The multi-level counter supports both standing adult transactions and accessible, child-height interactions, ensuring every student feels seen and included.

• **Tactile & Acoustic Detailing:** The integrated vertical wood slats and diagonal panels provide a natural, tactile texture that diffuses sound, helping to maintain a quiet and calm auditory environment.

• **Safety-First Geometry:** Every edge is designed with a significant radius to eliminate sharp corners, prioritizing the physical safety of students during high-energy transitions.

• **Structured Organization:** The integrated privacy cowl and cable management systems keep administrative clutter and wiring hidden, promoting a clean, distraction-free visual environment.

• **Sensory-Neutral Palette:** The use of matte, non-reflective surfaces combined with natural wood tones reduces visual glare and provides a grounding, low-arousal aesthetic for neurodiverse individuals.



Baltic Birch Plywood



A36 Carbon Steel Plate



Solid Ash Hardwood



Matte Anti-Fingerprint Laminate



Natural Oak Veneer



Acoustic PET Felt

• **Ergonomic Geometry:** The bold, triangular backrest provides a focused lumbar support point while its wide upper taper allows for lateral shoulder movement, balancing rigid form with physical freedom.

• **Structural Hybrid Joinery:** The connection between the backrest and seat utilizes a hidden steel spline system, ensuring that the avant-garde "point-contact" aesthetic remains structurally sound and safe for long-term use.

• **Fluid-to-Linear Transition:** The contrast between the sharp, geometric front legs and the continuous, laminated "U-shaped" rear support serves as a visual metaphor for stability and creative flow.

• **Biophilic Materiality:** Constructed from a combination of solid Ash and Baltic Birch, the chair offers a grounding, tactile experience that brings the warmth of the natural world into the learning or office environment.

• **Neuro-Inclusive Stability:** The splayed leg architecture creates a wide, non-tip footprint, providing a sense of physical security and "groundedness" for individuals with sensory processing or balance needs.

• **Visual Clarity & Low Arousal:** The use of matte-finished natural wood tones and clean, intentional lines reduces visual noise, contributing to a calm and distraction-free workspace.

10. CONCLUSION

The design of an autism school requires a thoughtful and user-centered approach that responds to the unique needs of children with autism. Through detailed study, case analysis, and design development, it is understood that the built environment plays a significant role in influencing behavior, learning ability, and emotional well-being.

A well-designed space must provide a structured, predictable, and sensory-friendly environment that reduces anxiety and supports focus and independence.

The project emphasizes the importance of clear spatial organization, simple circulation, and proper zoning, which help children navigate spaces easily and comfortably. The use of appropriate materials, soft color palettes, controlled lighting, and acoustic treatments contributes to creating a low-arousal environment that minimizes sensory overload.

Integration of therapy spaces, learning areas, and recreational zones ensures a balance between education, development, and relaxation.

Additionally, the inclusion of nature and outdoor spaces enhances sensory experiences and promotes emotional stability. The design also considers the needs of teachers, therapists, and caregivers, ensuring functionality, safety, and ease of supervision.

Overall, the project demonstrates how design can act as a powerful tool in improving the quality of life for children with autism by creating a safe, inclusive, and supportive environment.

In conclusion, the autism school is not just an educational space, but a carefully designed environment that fosters learning, independence, social interaction, and holistic development, contributing positively to the growth and well-being of its users.

11.REFERENCE

<https://www.autismspeaks.org>

<https://www.autism.org.uk>

<https://www.autism-india.org>

<https://www.vexcel.org>

<https://www.ummeed.org>

<https://www.necc.org>

<https://www.archdaily.com>

<https://www.dezeen.com>

<https://cae.org.uk>

<https://www.who.int>

<https://www.dimensions.com/element/classroomlayouts>

<https://www.dimensions.com/element/school-corridor-widths>